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CS 307

Operating Systems

Faculty Faculty of Engineering and Natural Sciences

Semester Fall 2025-2026

Course CS 307 - Operating Systems

Time/Place	<u>Time</u>	<u>Week Day</u>	<u>Place</u>	<u>Date</u>
	10:40-11:30	Wed	FENS-G077	Sep 29, 2025-Jan 3, 2026
	08:40-10:30	Thu	SBS-G071	Sep 29, 2025-Jan 3, 2026

Level of course Undergraduate

Course Credits SU Credit:3, ECTS:6, Engineering:6

Prerequisites CS 204

Corequisites CS 307R

Course Type Lecture

Instructor(s) Information

Süha Orhun Mutluergil

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Course Information

Catalog Course Description

This course covers fundamental aspects of operating systems: management and virtualization of resources such as CPU, memory space and peripheral devices. Topics include concurrent processes, mutual exclusion, process communication, cooperation, deadlocks, semaphores, scheduling, protection, memory management, file system abstraction and persistent memory organization. The course will also highlight important aspects of operating systems such as UNIX and Windows.

Learning Outcomes:

1.	To have a high-level understanding of computer systems, how an operating system works, and the general structure of an operating system.
2.	To have an understanding of how OS handles multiple processes, and multi-threading including synchronization, scheduling, and memory management.
3.	To be aware of some common problems due to interprocess communication, how these problems are modeled and solved.
4.	To have an understanding of how file systems are implemented and how I/O operations are performed.
5.	To have an understanding of basics of threats to computer security

Course Objective

To supply the students with the theory and practice of operating systems and operating system design principles.

Sustainable Development Goals (SDGs) Related to This Course:

 7 AFFORDABLE AND CLEAN ENERGY	Affordable and Clean Energy
 9 INDUSTRY, INNOVATION AND INFRASTRUCTURE	Industry, Innovation and Infrastructure

Course Materials

Resources:

Primary Textbook: Remzi H. Arpaci-Dusseau and Andrea C. Arpaci-Dusseau. Operating Systems: Three Easy Pieces, 1.00 Edition, Published by CreateSpace Independent Publishing Platform, 2018, ISBN: 978-1985086593. Free online access: <https://pages.cs.wisc.edu/~remzi/OSTEP/>

Technology Requirements:

CS307 server for programming assignments:
cs307.sabanciuniv.edu OR 10.3.0.91

Login with your SU username and password.
It will be available after add/drop period.

Policies

SU Academic Integrity Statement:

<https://www.sabanciuniv.edu/en/academic-integrity-statement>